

Sharp time decay at the bottom of the spherical spectrum

A full answer to the first sharpness conjecture in arXiv:2504.21479

Automated mathematics discovery run `fa_banach_001`

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Status: candidate full solution, likely valid. The two conjectured pointwise powers are attained with explicit nonzero leading coefficients. This packet does not address the separate conjecture about the L^1 growth of the distinguished-Laplacian kernel.

1 Source conjecture

Kuznetsova and Song study oscillatory functions of shifted Laplacians on a noncompact symmetric space G/K [1]. On page 3, after comparing their gapless $|t|^{-\nu}$ pointwise estimate with the positive-gap $|t|^{-\nu/2}$ estimate of Anker and Zhang [2], they write:

“Note the difference in the exponent between the two cases, ν against $\nu/2$. We conjecture that both are optimal, for small x^+ , and reflect different behaviour of the two operators.”

The complete source context is reproduced below.

(1) If $p = \nu$, then for every $x \in G$ and $t \neq 0$,

$$|k_t(x)| \lesssim |t|^{-\nu} (1 + |x^+|)^{\nu+d} e^{-\rho(x^+)}.$$

(2) If $p = \lfloor l/2 \rfloor$, then for t large enough and $|x^+| \leq 3|t|$,

$$|k_t(x)| \lesssim |t|^{(1-l)/2} (\log |t|)^d e^{-\rho(x^+)}.$$

(3) If $p = 2d + \lfloor l/2 \rfloor$, then for $|t|$ large enough and $|x^+| > 3|t|$

$$|k_t(x)| \lesssim |x^+|^{-d-l} e^{-\rho(x^+)}.$$

(4) There is a constant δ depending on the group only such that for $|t|$ large enough and x with $|x^+| > \log |t|$ and $||t| - |x^+|| > \delta \log |t|$

$$|k_t(x)| \lesssim (|x^+|^{-d-b-l} + |x^+|^{(1-l)/2} ||x^+| - |t||^{-d-b-(l+1)/2}) \times (\log |t|)^{2(2d+b+l)+1} e^{-\rho(x^+)}.$$

The first estimate applies also to small t , though our main interest is in large time t .

Statement 1 of this theorem is analogous to (1.2) above. Note the difference in the exponent between the two cases, ν against $\nu/2$. We conjecture that both are optimal, for small x^+ , and reflect different behaviour of the two operators.

We prove locally uniform asymptotic formulas on every compact spatial set. In particular, both upper-bound powers are optimal throughout the small x^+ regime.

2 Setting and statement

Let G be a connected noncompact semisimple Lie group with finite center, K a maximal compact subgroup, and $\mathfrak{a}$ a Cartan subspace of real dimension l . Let Σ_r^+ be the set of positive reduced roots, $d = |\Sigma_r^+|$, and

$$\nu = 2d + l.$$

Write φ_λ for the elementary spherical functions and $\mathbf{c}(\lambda)$ for the Harish-Chandra c -function, using the normalization of [1]. Spherical inversion gives the gapless kernel

$$k_t^0(x) = C_G \int_{\mathfrak{a}^*} e^{it|\lambda|} \psi(|\lambda|) \varphi_\lambda(x) |\mathbf{c}(\lambda)|^{-2} d\lambda. \quad (1)$$

Here $C_G \neq 0$ is the inversion constant.

For each $\alpha \in \Sigma_r^+$ put

$$a_\alpha = \frac{m_\alpha}{4} + \frac{1}{2}, \quad b_\alpha = \frac{m_\alpha}{4} + \frac{m_{2\alpha}}{2}, \quad \alpha_0 = \frac{\alpha}{\langle \alpha, \alpha \rangle}.$$

If c_0 is the constant in the Gindikin–Karpelevich formula, define

$$D = |c_0|^{-2} \prod_{\alpha \in \Sigma_r^+} \Gamma(a_\alpha)^2 \Gamma(b_\alpha)^2, \quad A_{\mathbf{c}} = D \int_{\mathbb{S}^{l-1}} \prod_{\alpha \in \Sigma_r^+} \langle \omega, \alpha_0 \rangle^2 d\omega. \quad (2)$$

The integrand is nonnegative and is positive away from finitely many great subspheres, so $A_{\mathbf{c}} > 0$.

Theorem 1 (Sharp gapless and positive-gap asymptotics). *Let $Q \subset G$ be compact and let $t \rightarrow +\infty$.*

1. *If ψ is a Schwartz function on $[0, \infty)$ and $\psi(0) \neq 0$, then, uniformly for $x \in Q$,*

$$k_t^0(x) = C_G A_{\mathbf{c}} \varphi_0(x) \psi(0) \Gamma(\nu) e^{i\pi\nu/2} t^{-\nu} + o_Q(t^{-\nu}). \quad (3)$$

2. *Let $\varepsilon > 0$. Let m be smooth, compactly supported, even near zero, and satisfy $m(0) \neq 0$. For*

$$k_{t,\varepsilon}(x) = C_G \int_{\mathfrak{a}^*} e^{it\sqrt{|\lambda|^2 + \varepsilon}} m(|\lambda|) \varphi_\lambda(x) |\mathbf{c}(\lambda)|^{-2} d\lambda, \quad (4)$$

one has, uniformly for $x \in Q$,

$$\begin{aligned} k_{t,\varepsilon}(x) &= C_G A_{\mathbf{c}} \varphi_0(x) m(0) 2^{\nu/2-1} \varepsilon^{\nu/4} \Gamma(\nu/2) \\ &\quad \times e^{it\sqrt{\varepsilon} + i\pi\nu/4} t^{-\nu/2} + o_Q(t^{-\nu/2}). \end{aligned} \quad (5)$$

3. *Formula (5) remains valid for the uncut Bessel amplitude*

$$m(r) = (r^2 + \varepsilon)^{-\alpha/2}, \quad \operatorname{Re} \alpha > n,$$

where $n = \dim(G/K)$; in this case $m(0) = \varepsilon^{-\alpha/2}$.

Since φ_0 is positive and has a positive minimum on Q , both leading coefficients are bounded away from zero in modulus on Q . Consequently, for all sufficiently large t ,

$$\inf_{x \in Q} |k_t^0(x)| \geq c_Q t^{-\nu}, \quad \inf_{x \in Q} |k_{t,\varepsilon}(x)| \geq c_{Q,\varepsilon} t^{-\nu/2}.$$

Thus neither power can be uniformly improved on any compact spatial region.

3 Proof intuition

The time powers come entirely from the bottom of the spherical spectrum. Every reduced positive root contributes an exact squared linear zero to the Plancherel density. There are d such roots; together with the l -dimensional polar Jacobian this produces a radial amplitude of exact order $r^{2d+l-1} = r^{\nu-1}$. With no spectral gap, the phase is tr , so the half-line endpoint Fourier lemma gives $t^{-\nu}$.

With a positive gap, the phase has a quadratic stationary endpoint: $\sqrt{r^2 + \varepsilon} = \sqrt{\varepsilon} + O(r^2)$. The exact variable $s = \sqrt{r^2 + \varepsilon} - \sqrt{\varepsilon}$ converts it to a linear phase, and the transformed amplitude has order $s^{\nu/2-1}$. This gives $t^{-\nu/2}$. The angular coefficient is an integral of a nonzero square, so neither leading term can cancel.

4 Proof

4.1 Exact order of the Plancherel density

The Gindikin–Karpelevich product in [1, Eq. (2.6)] is

$$\mathbf{c}(\lambda) = c_0 \prod_{\alpha \in \Sigma_r^+} \frac{2^{-iu_\alpha} \Gamma(iu_\alpha)}{\Gamma(a_\alpha + iu_\alpha/2) \Gamma(b_\alpha + iu_\alpha/2)}, \quad u_\alpha = \langle \lambda, \alpha_0 \rangle.$$

For real u ,

$$\frac{|\Gamma(a + iu/2)|^2 |\Gamma(b + iu/2)|^2}{|\Gamma(iu)|^2} = \Gamma(a)^2 \Gamma(b)^2 u^2 h_{a,b}(u),$$

where $h_{a,b}$ is positive, even, and real analytic near zero, with $h_{a,b}(0) = 1$. This follows from $\Gamma(iu)^{-1} = iu + O(u^2)$; evenness removes the odd term after taking modulus square. Hence, uniformly for $\omega \in \mathbb{S}^{l-1}$,

$$|\mathbf{c}(r\omega)|^{-2} = Dr^{2d} \prod_{\alpha \in \Sigma_r^+} \langle \omega, \alpha_0 \rangle^2 (1 + O(r^2)). \quad (6)$$

This identity remains uniform at root hyperplanes because the squared root forms have been factored out exactly.

For $r \geq 0$ define the polar density

$$q_x(r) = r^{l-1} \int_{\mathbb{S}^{l-1}} \varphi_{r\omega}(x) |\mathbf{c}(r\omega)|^{-2} d\omega.$$

Spherical functions are analytic in λ and $\varphi_{r\omega}(x) = \varphi_0(x) + O_Q(r)$ uniformly for $x \in Q$. The term linear in r is odd in ω , whereas the leading product in (6) is even. Its angular integral vanishes. Therefore

$$q_x(r) = A_{\mathbf{c}} \varphi_0(x) r^{\nu-1} + O_Q(r^{\nu+1}). \quad (7)$$

More precisely, $q_x(r) = r^{\nu-1} H_x(r^2)$ near zero, with H_x smooth and $H_x(0) = A_{\mathbf{c}} \varphi_0(x)$, locally uniformly in x .

4.2 Endpoint Fourier lemma

We use the standard half-line endpoint formula: if $\mu > 0$ and $g(s) = s^{\mu-1} G(s)$ is compactly supported on $[0, \infty)$, with G smooth near zero, then

$$\int_0^\infty e^{its} g(s) ds = G(0) e^{i\pi\mu/2} \Gamma(\mu) t^{-\mu} + o(t^{-\mu}). \quad (8)$$

The same conclusion holds after adding a tail that is $O(t^{-N})$ for some $N > \mu$. One proof inserts an Abel factor, evaluates $\int_0^\infty e^{-(\eta-it)s} s^{\mu-1} ds = \Gamma(\mu) (\eta - it)^{-\mu}$, subtracts the leading Taylor term of G , and integrates the remainder by parts. This also proves uniformity when G ranges over a bounded family of smooth functions.

4.3 Gapless asymptotic

Polar coordinates in (1) give

$$k_t^0(x) = C_G \int_0^\infty e^{itr} \psi(r) q_x(r) dr.$$

By (7), the endpoint amplitude is $A_{\mathbf{c}} \varphi_0(x) \psi(0) r^{\nu-1} + O_Q(r^\nu)$. Apply (8) with $\mu = \nu$. Away from zero the phase derivative is one. The derivative bounds for $\mathbf{c}^{-1}$ and φ_λ stated in [1, Sec. 2.2] give polynomial growth for all derivatives of q_x , uniformly on Q ; the Schwartz decay of ψ permits arbitrarily many integrations by parts. This proves (3).

4.4 Positive-gap asymptotic

Set $a = \sqrt{\varepsilon}$ and

$$s = \sqrt{r^2 + \varepsilon} - a.$$

Then

$$r^2 = s(s + 2a), \quad \frac{dr}{ds} = \frac{s + a}{r}. \quad (9)$$

Using (7), the transformed leading amplitude is

$$\begin{aligned} q_x(r)m(r)\frac{dr}{ds} &= A_{\mathbf{c}}\varphi_0(x)m(0)(s+a)\{s(s+2a)\}^{(\nu-2)/2} + o_Q(s^{\nu/2-1}) \\ &= A_{\mathbf{c}}\varphi_0(x)m(0)2^{\nu/2-1}\varepsilon^{\nu/4}s^{\nu/2-1} + o_Q(s^{\nu/2-1}). \end{aligned}$$

The phase is now $e^{ita}e^{its}$. Formula (8) with $\mu = \nu/2$ proves (5) for compactly supported m .

For $m(r) = (r^2 + \varepsilon)^{-\alpha/2}$, split with a cutoff supported near zero. The local calculation is unchanged and $m(0) = \varepsilon^{-\alpha/2}$. On the complementary region, $r/\sqrt{r^2 + \varepsilon}$ is bounded below. The derivative estimate for $\mathbf{c}^{-1}$ in [1, Theorem 2.4], together with the spherical-function derivative estimate, gives for every fixed j

$$\partial_r^j q_x(r) = O_Q((1+r)^{n-1}).$$

All derivatives of the Bessel amplitude are bounded by a constant times $(1+r)^{-\operatorname{Re} \alpha}$ at worst. Thus every term arising from repeated nonstationary integration by parts is integrable when $\operatorname{Re} \alpha > n$. The tail is $O_Q(t^{-N})$ for every N , and the proof is complete. $\square$

5 Verification, scope, and novelty

The algebraic checker in `code/check_wave_endpoint_constants.py` verifies exact model Fourier integrals, the coefficient in (9), and positivity for an A_2 root-system model. These checks do not replace the proof.

This theorem proves pointwise kernel sharpness. It does not assert that every finite- p dispersive operator norm has a matching lower bound. The condition $\psi(0) \neq 0$ is essential to the stated exponent: if the amplitude vanishes at zero, the first nonzero Taylor term gives a higher endpoint order. The separate conjecture in [1] concerning the optimal L^1 growth exponent $(\nu - 1)/2$ for the distinguished Laplacian remains open.

Before promotion, the run’s cheap indexes were searched by arXiv id, title, exact conjecture sentence, and the terms “wave kernel”, “spectral gap”, and “sharp time decay”. A bounded primary arXiv search covered arXiv:2504.21479, the exact phrase, the authors, and the Anker–Zhang antecedent arXiv:2010.08467. No later paper claiming this endpoint asymptotic or resolving the conjecture was found. This is not an exhaustive novelty certification.

Human review recommendation. Send to a specialist in spherical harmonic analysis. The two highest-value checks are the uniform analytic factorization in (6) across root hyperplanes and the stated $\operatorname{Re} \alpha > n$ tail threshold.

References

- [1] Y. Kuznetsova and Z. Song, *Shifted wave equation on noncompact symmetric spaces*, arXiv:2504.21479v2, 2026.
- [2] J.-Ph. Anker and H.-W. Zhang, *Wave equation on general noncompact symmetric spaces*, American Journal of Mathematics 146 (2024), 983–1031; arXiv:2010.08467.